

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 11A.2

RATE EQUATIONS, RATE CONSTANTS AND ORDERS OF REACTION

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A.2: Rate Equations, Rate Constants and Orders of Reaction
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	4 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 11A.2

Q1. Q1. The reaction between NO and O₂ is second order with respect to NO and first order with respect to O₂. Write the rate equation for this reaction.

[1 Marks]**WCH14/01/Oct22/Q2(a)**

Q2. Q2. Deduce the units of the rate constant, k, for the reaction in Q1.

[1 Marks]**WCH14/01/Oct22/Q2(b)**

Q3. Q3. What is meant by the term "order of reaction" with respect to a given reactant?

[1 Marks]**WCH14/01/Jun21/Q1(a)**

Q4. Q4. Explain what is meant by the overall order of a reaction.

[1 Marks]**WCH14/01/Jun21/Q1(b)**

SECTION B: WORKED MARK SCHEME — 11A.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Q1. The reaction between NO and O ₂ is second order with respect to NO and first order with respect to O ₂ . Write the rate equation for this reaction. • $\text{rate} = k[\text{NO}]^2[\text{O}_2]$ (1)	[1]
Q2	Q2: Q2. Deduce the units of the rate constant, k, for the reaction in Q1. • $\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$ (1)	[1]
Q3	Q3: Q3. What is meant by the term "order of reaction" with respect to a given reactant? • The power to which the concentration of that reactant is raised in the rate equation (1)	[1]
Q4	Q4: Q4. Explain what is meant by the overall order of a reaction. • The sum of the powers of the concentration terms in the rate equation (1)	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 11A.2**FAQ #1: How do I determine units for k?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How do I determine units for k?

MODEL ANSWER / EXAMINER EXPECTATION:

Substitute the units of rate ($\text{mol dm}^{-3} \text{ s}^{-1}$) and concentrations (mol dm^{-3}) into the rate equation and cancel them out.

FAQ #2: Is zero order included in the rate equation?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Is zero order included in the rate equation?

MODEL ANSWER / EXAMINER EXPECTATION:

No, any concentration raised to the power of zero is 1, so it is typically omitted from the rate equation.

FAQ #3: Can orders of reaction be fractional?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Can orders of reaction be fractional?

MODEL ANSWER / EXAMINER EXPECTATION:

Yes, though they are usually integers (0, 1, 2) in A-level questions, fractional orders are possible.